DO NOW · AO1 REVISION

Resistance and I–V characteristics (RP)

Electricity
Paper 1 · Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 8 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the two measurements needed to find the resistance of a component.**	The potential difference across it and the current through it	☐
2	**Wire length and resistance: name the independent variable.**	Length of the wire	☐
3	**Wire length and resistance: name the dependent variable.**	Resistance of the wire	☐
4 ★	**Wire length and resistance: name three control variables.**	The temperature, thickness and material of the wire	☐
5 ★	**Describe how the length of a wire affects its resistance.**	The resistance increases (it is directly proportional to length).	☐
6 ★	**In the wire-length practical, the fixed crocodile clip is at the 1 cm mark, but every length is read from 0 cm. Name the type of error this causes.**	A zero error (a systematic error): every length is 1 cm too long.	☐
7 ★	**Explain why the potential difference is kept small when measuring the resistance of a wire.**	So the current is small and the wire does not heat up.	☐
8	**What does it mean for two resistors to be in series?**	They are connected one after the other, on a single path.	☐
9	**What does it mean for two resistors to be in parallel?**	They are on separate branches.	☐
10 ★	**Describe how to calculate the total resistance of two resistors in series.**	Add the two resistances: $R_{total} = R_1 + R_2$.	☐

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	QUESTION	ANSWER	✓
11	**A lamp, a power supply and a variable resistor are in series. State two ways to increase the potential difference across the lamp.**	Increase the supply potential difference. Decrease the resistance of the variable resistor.	☐
12 ★	**Finding the I–V characteristic of a component: name the independent and dependent variables.**	Independent: potential difference across the component Dependent: current through it	☐
13 ★	**Describe the graph of resistance against length for a wire.**	A straight line through the origin.	☐

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